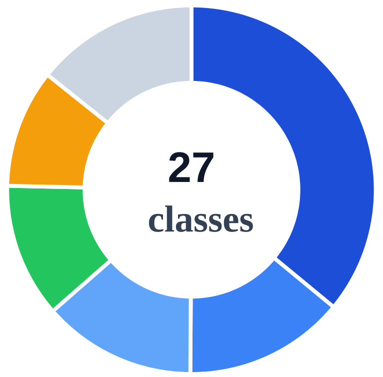
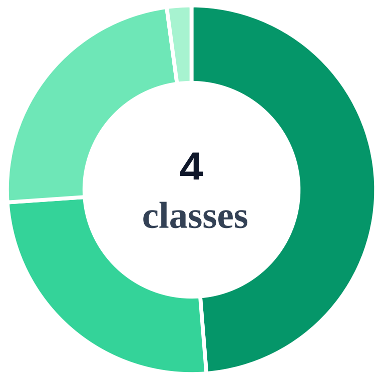
≈100K	30 s	≈3.0M	833.6 h	3,294.7 km
non-overlapping clips	per navigation clip	benchmark frames	navigation data	traveled distance

A. 3D Object Annotation Composition



🟦	Pedestrian	35.98%
🟦	Scooter	14.12%
🟦	Car	13.46%
🟩	Cone	11.78%
🟧	Bicycle	10.38%
⬜	Other 22 classes	14.28%

B. Road-Line Annotation Composition



🟩	sidewalk_curb	48.71%
🟩	road_curb	25.20%
🟩	dividing_line	23.94%
🟩	crosswalk	2.15%

C. Instruction Duration Distribution

follow_path	36.16%
continue_straight	15.15%
stationary	13.55%
take_left_branch	10.93%
take_right_branch	10.74%
obstacle_avoid_left	10.40%
others	3.07%

D. Instruction Coverage and Split Protocol

Primary navigation	100.00%
Special motion	35.91%
Obstacle interaction	22.42%

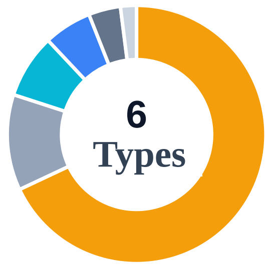
Primary navigation

Special motion | Obstacle interaction

E. Fine-Grained Scene Distribution

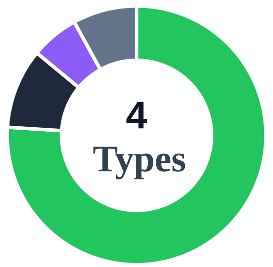
Sidewalk	25.00%
Street	13.00%
Campus	10.00%
Park	9.00%
Cross intersection	8.00%
Residential area	8.00%
Open ground	7.00%
Narrow passage	6.00%
Others	14.00%

F. Environment and Time Conditions



Weather

🟧	Sunny	68%
⬜	Cloudy	12%
🟦	Windy	8%
🟦	Rainy	6%
⬛	Overcast	4%
⬜	Other	2%



Time

🟩	Daytime	76%
⬛	Night	10%
🟪	Dusk/Dawn	6%
⬛	Low-light shade	8%

G. Spatial Complexity Around the Robot

Object distance distribution

0–20 m	76.52%
20–50 m	22.94%
> 50 m	0.54%

Objects per frame

1–5	61.75%
6–10	18.36%
11–20	12.57%
> 20	6.63%

Mean 13.93 m | P50 11.10 m | P95 34.83 m | P99 46.71 m

Mean: 6.94 / frame

H. Ego-Motion Statistics

Speed

Mean	1.061 m/s
P50	1.235 m/s
P95	1.351 m/s
1–1.5 m/s	72.89%
< 0.1 m/s	5.84%

Absolute curvature

P50	0.0391 1/m
P95	0.4194 1/m
P99	0.8188 1/m
56.45%	<0.05 1/m